

HabitSnap - A Streak-Based Habit Tracking Mobile Application

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ABSTRACT

HabitSnap is a mobile-based application designed to help users build and maintain positive daily habits through a simple and engaging interface. In today's fast-paced lifestyle, individuals often struggle to stay consistent with their personal goals such as exercising, studying, or maintaining healthy routines. HabitSnap addresses this problem by providing an easy-to-use platform where users can create, track, and monitor their habits efficiently.

The application allows users to add custom habits, mark daily progress, and visualize their consistency over time. It incorporates features such as reminders, progress tracking, and intuitive dashboards to motivate users and improve productivity. By presenting habit data in a structured and visually appealing manner, users can easily identify patterns and stay committed to their goals.

Developed using modern technologies, HabitSnap ensures smooth performance and user-friendly interaction. The system emphasizes simplicity, accessibility, and effectiveness, making it suitable for beginners as well as regular users. Overall, HabitSnap serves as a practical tool for self-improvement by encouraging discipline, consistency, and long-term habit formation.

The system enables users to create personalized habits such as fitness activities, study goals, or daily routines, and track their progress on a day-to-day basis. It includes key features like habit reminders, streak tracking, and visual progress representation through charts or summaries. These features help users stay motivated by providing clear insights into their performance and consistency over time.

HabitSnap is designed with a clean and user-friendly interface to ensure ease of use for all types of users, including beginners. The backend system efficiently manages user data and habit records, ensuring reliable storage and retrieval of information. The application focuses on improving user productivity and self-discipline by encouraging consistency and accountability.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

HabitSnap is a mobile-based habit tracking application developed for Android using modern development tools and technologies. The application is designed to help users build, monitor, and maintain positive daily habits in a simple and structured way. In today's fast-paced environment, individuals often struggle to stay consistent with their goals such as exercising, studying, or maintaining a healthy lifestyle. HabitSnap addresses this challenge by providing a centralized platform to track and improve daily routines.

The application enables users to create personalized habits and record their daily progress. It includes features such as habit tracking, streak monitoring, reminders, and progress visualization. All data is stored locally on the device using efficient storage mechanisms, ensuring fast performance and user privacy without relying on internet connectivity.

The user interface is designed using modern layouts to provide a clean, responsive, and user-friendly experience across different devices. Each feature of the application is built to work seamlessly, allowing users to easily navigate, update habits, and monitor their improvement over time. HabitSnap acts as a digital companion that encourages consistency and supports long-term personal development.

1.2 SCOPE OF THE WORK

The scope of this project focuses on the design and development of HabitSnap, a lightweight and efficient habit tracking application for Android devices. The system aims to provide essential tools required for habit formation and monitoring within a single platform.

The application allows users to create, edit, and delete habits, track their daily completion status, and view their progress over time. Features such as streak tracking and reminders help improve user engagement and consistency. The system is built using Android development frameworks and utilizes local storage techniques to manage user data securely and efficiently.

In addition, the project emphasizes a user-friendly interface with responsive design elements to ensure accessibility across various screen sizes. Performance optimization and simplicity are key considerations, making the application suitable for all types of users, including beginners.

The project does not rely on cloud integration, focusing instead on offline functionality, which ensures data privacy and reduces dependency on external services. Overall, the scope is to develop a reliable and easy-to-use application that promotes habit formation and improves productivity.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim of this project is to design and develop HabitSnap, an Android-based application that helps users build and maintain positive habits through consistent tracking and monitoring. The application aims to simplify the process of habit formation by providing an organized and user-friendly platform.

The objectives of the project include:

- To develop a system that allows users to create and manage daily habits efficiently.
- To implement a habit tracking mechanism that records daily progress.
- To provide visual insights such as streaks and progress summaries to motivate users.
- To design a simple and intuitive user interface for better usability.
- To ensure data privacy by storing all user information locally on the device.
- To build a lightweight and fast-performing application suitable for mobile devices.

CHAPTER 2

LITERATURE SURVEY

Recent research on habit tracking systems highlights the importance of consistency and user engagement in building productive routines. A mobile habit tracking application using local storage and progress visualization is presented in [1]. The system allows users to create daily habits, track completion, and monitor streaks. It uses lightweight data storage techniques such as Shared Preferences to ensure offline functionality. A user study reported improved consistency and a noticeable increase in habit retention. This study supports the core habit tracking functionality of HabitSnap.

A study on reminder-based productivity applications is discussed in [2], where timely notifications are used to improve user adherence to daily tasks and habits. The system utilizes Android's notification services to send alerts and reminders at scheduled intervals. Experimental results showed that users who received reminders were significantly more consistent in completing their tasks. This research supports the reminder feature implemented in HabitSnap.

Another research work focusing on data visualization in mobile applications is presented in [3]. The application provides graphical representations of user progress through charts and summaries, helping users understand their performance over time. The study concluded that visual feedback increases user motivation and engagement. This concept is applied in HabitSnap through progress tracking and streak visualization features.

A habit formation study using behavioral psychology principles is explained in [4]. The system emphasizes the importance of small, consistent actions and tracking progress over time to build long-term habits. It highlights that tracking daily performance and maintaining streaks significantly improves habit formation success rates. This research forms the conceptual foundation for the design of HabitSnap.

Additionally, a lightweight offline-first mobile application architecture is discussed in [5], where applications rely entirely on local storage for faster performance and enhanced privacy. The study shows that eliminating cloud dependency reduces latency and improves user trust. This approach is implemented in HabitSnap by storing all user data locally on the device.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

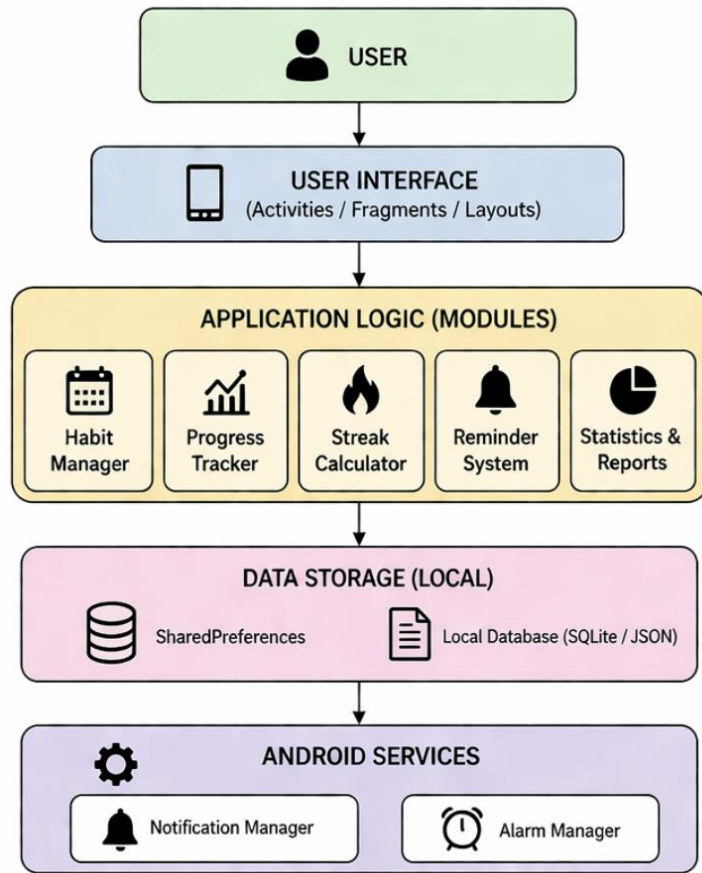


Fig 3.1.1 – Architecture Diagram

The HabitSnap application follows a modular and lightweight architecture designed for efficient performance and ease of use. The system is structured into different layers, including the User Interface (UI), Application Logic, and Data Storage layer. Each component works together to provide a smooth and responsive user experience.

The core functionalities such as Habit Management, Progress Tracking, Streak Calculation, and Reminder System are implemented as independent modules. These modules interact with each other through well-defined interfaces, ensuring flexibility and scalability of the application.

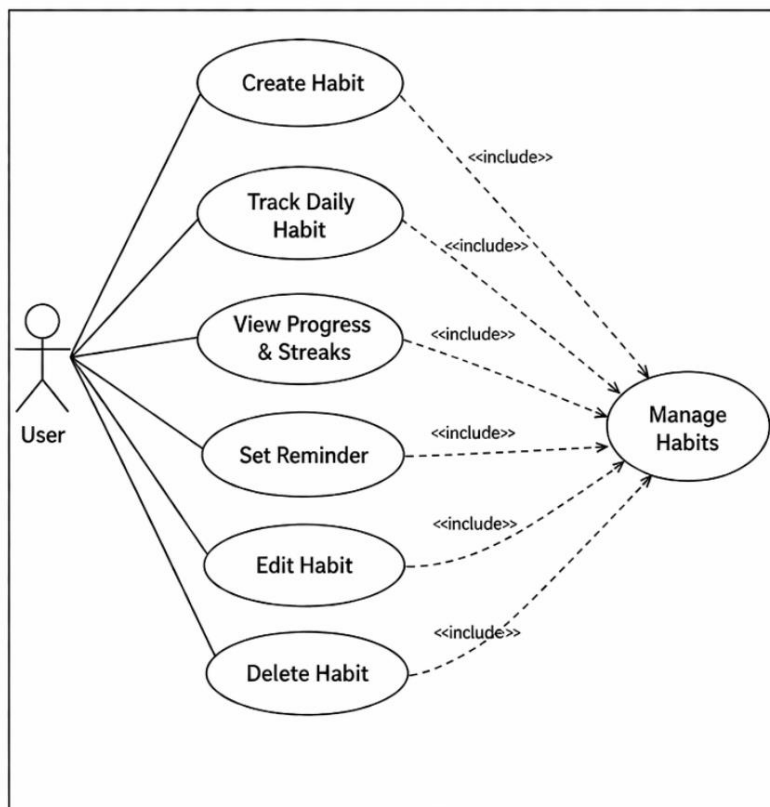
Data is stored locally on the device using storage mechanisms such as Shared Preferences or lightweight databases, allowing fast access and ensuring user privacy.

without requiring internet connectivity. The architecture avoids heavy dependencies and focuses on simplicity, making the application efficient even on low-end devices.

The system also integrates Android's native components such as notification services for reminders and lifecycle management for smooth operation. This modular design allows easy maintenance and future enhancements.

3.2 USE CASE DIAGRAM

Fig 3.2.1 – Use Case Diagram



The Use Case Diagram of HabitSnap represents the interaction between the user and the system. The primary actor is the user, who interacts with various functionalities provided by the application.

The main use cases include creating habits, updating daily progress, viewing habit streaks, and managing existing habits. Users can add new habits, mark them as

completed each day, edit or delete habits, and monitor their progress through visual summaries.

Additionally, users can receive reminders to complete their habits, which helps in maintaining consistency. All these functionalities are designed to work together seamlessly, providing a complete habit tracking experience.

3.3 HARDWARE SPECIFICATIONS

The HabitSnap application is designed to run efficiently on standard Android smartphones without requiring specialized hardware. It ensures smooth performance even on devices with moderate specifications.

The recommended hardware specifications are:

- **Processor:** Quad-core processor (e.g., Snapdragon 660 or higher)
- **RAM:** Minimum 3 GB (Recommended: 4 GB for better performance)
- **Storage:** 32 GB internal storage or higher
- **Graphics:** Integrated graphics (no dedicated GPU required)
- **Display:** Touchscreen smartphone
- **Operating System:** Android 8.0 (Oreo) or higher

These specifications ensure that the application runs smoothly while handling habit tracking, reminders, and data storage efficiently.

3.4 SOFTWARE SPECIFICATIONS

The HabitSnap application is developed using modern Android development technologies to ensure performance, scalability, and ease of use.

- **Programming Language:** Kotlin
- **Development Environment:** Android Studio
- **UI Design:** XML Layouts / Jetpack Components

- **Design Principles:** Material Design for clean and user-friendly interface
- **Data Storage:**
 - Shared Preferences (for lightweight data storage)
 - Local database (SQLite / Room, if implemented)
- **Backend:**
 - No external backend (offline-first application)
- **Libraries & Tools:**
 - Android SDK
 - Notification Manager (for reminders)
 - Kotlin Standard Library

The software is designed to function completely offline, ensuring data privacy and fast performance. The modular structure allows for easy updates and future enhancements such as cloud integration or analytics features.

CHAPTER 4

PROPOSED SYSTEM

The proposed system, HabitSnap, is designed to provide a simple and effective platform for users to build and maintain daily habits. The application focuses on improving consistency and productivity by offering essential habit tracking features in a single, user-friendly interface.

The system enables users to create habits, track their daily completion, and monitor their progress over time. It is designed as an offline-first application, ensuring that all user data is stored locally on the device for faster access and enhanced privacy.

Key Components of the Proposed System:

- **Habit** **Manager:**
Allows users to create, update, and delete habits based on their personal goals.
- **Daily** **Tracking** **System:**
Enables users to mark habits as completed each day, helping maintain consistency.
- **Streak** **Tracker:**
Tracks continuous completion of habits and displays streak counts to motivate users.
- **Reminder** **System:**
Sends notifications to remind users to complete their habits on time.
- **Progress** **Visualization:**
Displays user performance using simple summaries or visual indicators to show improvement over time.

The system integrates these components into a cohesive workflow, ensuring smooth interaction between features. All data is stored using local storage mechanisms such as SharedPreferences or a lightweight database, eliminating the need for internet connectivity. The user interface is designed using modern Android UI components to provide a clean, responsive, and intuitive experience.

CHAPTER 5

MODULE DESCRIPTION

The HabitSnap application consists of several modules that work together to provide a complete habit tracking system. Each module is designed to perform a specific function while maintaining smooth interaction with other modules.

Habit Manager Module

This module allows users to create and manage their habits efficiently. Users can add new habits with titles and descriptions, edit existing habits, and delete habits when no longer needed. The module ensures proper organization of habits and provides an easy way to maintain daily routines.

Daily Tracking Module

The Daily Tracking module enables users to mark habits as completed on a daily basis. It records the user's activity and updates the status of each habit. This module plays a key role in maintaining consistency and tracking user engagement.

Streak Tracker Module

The Streak Tracker calculates and displays the number of consecutive days a habit has been completed. It motivates users to maintain their streaks and build long-term habits. Breaking a streak resets the count, encouraging consistent performance.

Reminder Module

This module allows users to set reminders for their habits. Notifications are triggered at specified times to ensure that users do not forget to complete their tasks. It uses Android notification services for efficient alert management.

Progress Visualization Module

The Progress Visualization module provides insights into user performance. It displays summaries such as completion rates and streak counts, helping users understand their progress over time. This visual feedback improves motivation and engagement.

KEY BENEFITS OF THE PROPOSED SYSTEM

Improved Habit Tracking

The application provides a structured and efficient way to track daily habits, helping users stay consistent and organized.

Increased Motivation

Features like streak tracking and progress visualization encourage users to maintain their habits and achieve their goals.

User-Friendly Interface

The simple and intuitive design makes the application easy to use for all types of users.

Offline Functionality

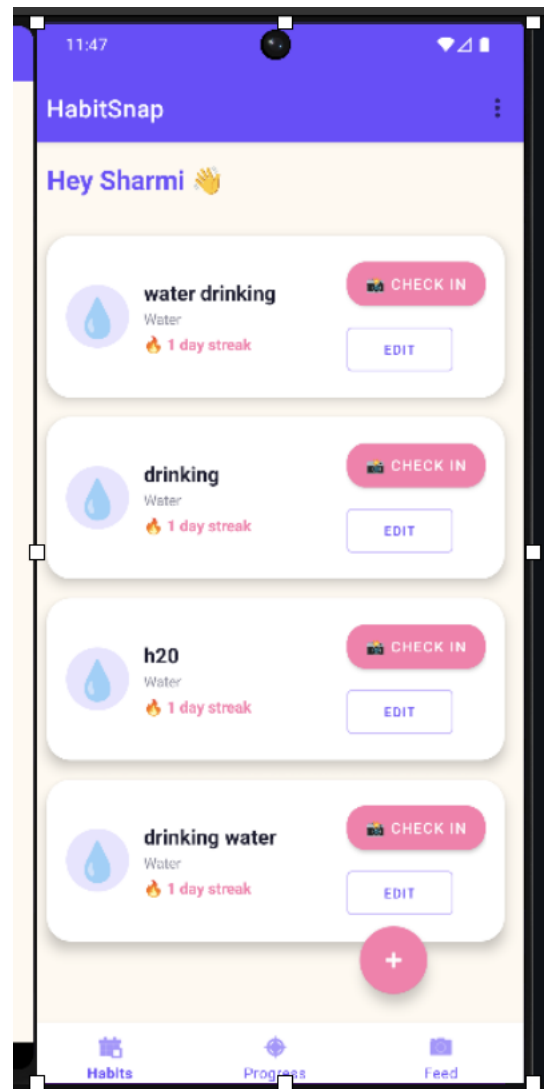
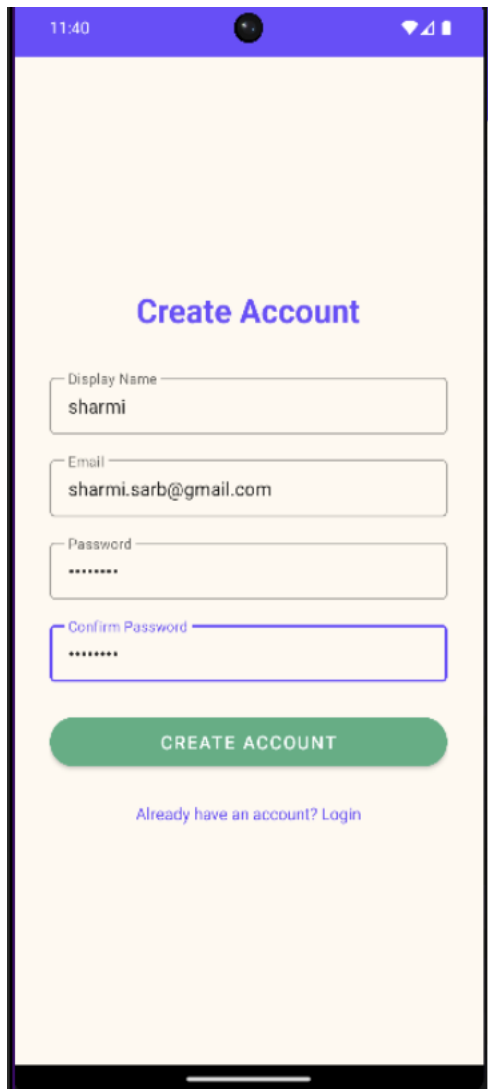
All data is stored locally, ensuring privacy and allowing the application to work without internet connectivity.

Lightweight and Efficient

The system is optimized for performance, making it suitable for devices with moderate specifications.

CHAPTER 6

OUTPUT



CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

HabitSnap provides a simple and effective solution for users to build and maintain positive daily habits. The application focuses on improving consistency, productivity, and self-discipline through structured habit tracking and progress monitoring. By offering features such as habit creation, daily tracking, streak

management, reminders, and progress visualization, the system helps users stay motivated and achieve their personal goals.

The application is designed with a user-friendly interface and lightweight architecture, ensuring smooth performance across different Android devices. Its offline-first approach enhances data privacy and allows users to access their information without relying on internet connectivity. Overall, HabitSnap acts as a reliable digital companion for personal development and habit formation.

Future Enhancements

- **User Personalization:**
Introducing customizable themes, habit categories, and flexible reminder settings to enhance user experience.
- **Cloud Backup and Sync:**
Adding cloud integration to allow users to back up their data and access it across multiple devices.
- **Advanced Analytics:**
Providing detailed insights such as weekly/monthly reports, habit success rates, and performance trends.
- **Gamification Features:**
Incorporating rewards, badges, and achievements to increase user engagement and motivation.
- **Cross-Platform Support:**
Expanding the application to platforms like iOS or web for a seamless multi-device experience.

CHAPTER 8

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